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Printed Pages – 6 **TEE – 402**

(Following Paper ID and Roll No. to be filled in your Answer Book)

PAPER ID : 2052 **Roll No.** | | | | | | | | | | |

B.Tech.

FOURTH SEMESTER EXAMINATION, 2005-2006

NETWORK ANALYSIS AND SYNTHESIS

Time : 3 Hours *Total Marks : 100*

Note : *(i) Attempt ALL questions.*

(ii) All questions carry equal marks.

(iii) In case of numerical problems assume data wherever not provided.

(iv) Be precise in your answer.

1. Attempt ***any three*** parts of the following : **($6\frac{2}{3} \times 3 = 20$)**

(a) Differentiate the terms "Cut-set" and "Tie-set" related to a linear graph.

Write the properties of incidence matrix and also obtain the incidence matrix of the graph shown in fig 1(a)

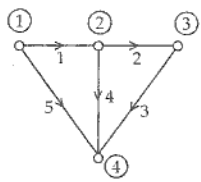


Fig 1(a)

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(b) In graph theory, derive the relation among submatrices of A, B and Q, where A, B and Q have their usual meanings.

(c) A resistive network is shown in fig 1(c), write corresponding tie-set matrix and obtain KVL equation using graph theory.

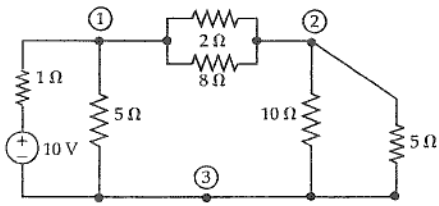


Fig. 1(c)

(d) Show the cut-sets for the graph of the network shown in fig 1(d) and develop the fundamental cut-set matrix.

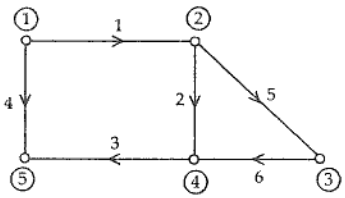


Fig. 1(d)

(e) Prove that given circuits shown in fig 1(e) (i) and 1e(ii) are duals of each other. Use graph theory only.

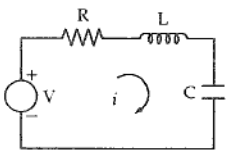


Fig. 1(e)(i)

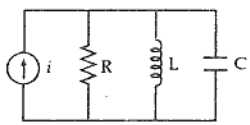


Fig. 1(e)(ii)

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2. Attempt *any three* parts of the following : **($6\frac{2}{3}$x3=20)**

(a) In the network shown in fig 2(a), find the current through capacitor using principle of superposition.

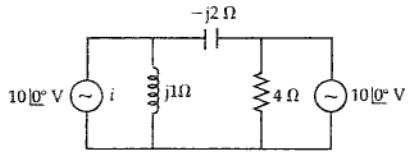


Fig. 2 (a)

(b) Find the Thevenin's equivalent circuit for the network given in fig 2(b) across A and B.

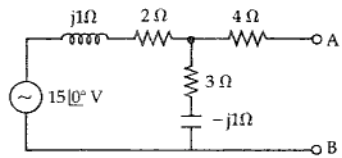


Fig. 2 (b)

(c) In the network shown in fig 2(c), find the value of R_L which results in maximum power transfer. Also find maximum power.

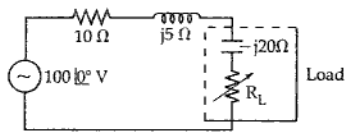


Fig. 2 (c)

(d) State the Tellegen's theorem and verify it for the network shown in fig 2(d).

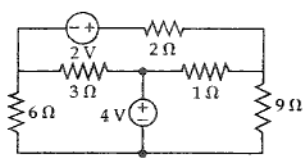


Fig. 2 (d)

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(e) Verify Reciprocity theorem of the network shown in fig 2 (e) and find V_x.

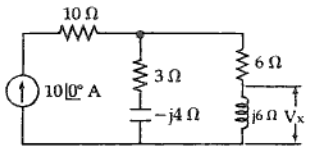


Fig. 2 (e)

3. Attempt ***any two*** parts of the following : **(10x2=20)**

(a) Enlist the necessary conditions for transfer function in terms of pole and zero locations. Draw the pole-zero diagram of the given function and obtain time domain response using if.

$$I(S) = \frac{2S}{(S+1)(S^2+2S+4)}$$

(b) (i) Using Routh - Hurwitz criterion, check the stability of the following polynomials.

$$P(S) = S^6 + S^5 + 3S^4 + 3S^3 + 3S^2 + 2S + 1$$

(ii) $P(S) = S^4 + 20KS^3 + 5S^2 + 10S + 15 = 0$

is a characteristic equation of control system. Find the range of K for the system to be stable.

(c) Draw the Bode plot of the following transfer function having unity feedback.

$$G(S) = \frac{50}{S(1+S)(5+S)(10+S)}$$

Find gain cross over and phase cross-over frequency.

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4. Attempt *any two* parts of the following : **(10x2=20)**

(a) Find Y-parameters of the network shown in fig 4(a) and check whether the circuit is symmetric and reciprocal or not.

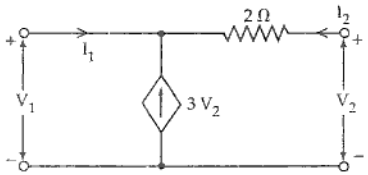


Fig. 4(a)

Also hence find h-parameters of the network.

(b) A 2-port network made up of passive linear resistors is fed at port 1 by an ideal voltage source of V volts. It is loaded at port 2 by a resistance R:

(i) With V = 10 volts and R = 6Ω, currents at ports 1 and 2 were 1.44 A and 0.2A respectively.

(ii) With V = 15 volt and R = 8Ω, the current at port 2 was 0.25 A. Determine π equivalent circuit of 2-port network.

(c) For the two part network shown in fig 4(c), find h parameters and hence find ABCD parameters.

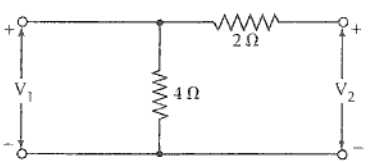


Fig. 4(c)

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5. Attempt *any three* parts of the following : $(6\frac{2}{3} \times 3 = 20)$

(a) Enlist the properties of an LC immittance driving point function. Obtain the driving point impedance in Laplace form of the given network across a-b shown in fig 5(a).

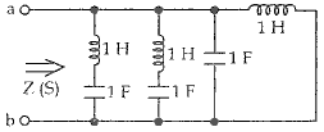


Fig. 5 (a)

(b) An impedance function is given by

$$Z(S) = \frac{2(S+1)(S+3)}{(S+2)(S+4)}$$

Find the R-L representation of Foster first form.

(c) List and explain the properties of a passive filter.

(d) A T-section low pass filter has series inductance 80 mH and shunt capacitance 0.022μF. Determine the cut-off frequency and nominal design impedance. Also design an equivalent π-section.

(e) A series LCR type band-stop filter has R = 1.5 kΩ; L = 140 mH and C = 300 pf. Find the

(i) Resonant frequency and

(ii) Bandwidth. Also find the cut-off frequencies.

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